Prem Raj

AI Full Stack Developer / Software Developer

New Delhi, India | +91 7726960080 | singhpremraj264@gmail.com | github.com/premrajsingh | linkedin.com/in/prem-raj | Portfolio

Summary

Final-year B.Tech Information Technology student graduating in 2027 with hands-on experience in React, React Native, Node.js, Express, FastAPI, MongoDB, MySQL, REST APIs, JWT authentication, and AI-powered RAG systems. Built and deployed full-stack products, automation tools, hostel management software, and multi-agent AI applications. Solved 250+ DSA problems.

Technical Skills

Languages:	JavaScript (ES6+), Python, C++, Java
Frontend:	React.js, React Native, Redux Toolkit, HTML5, CSS3, Tailwind CSS, Bootstrap, Vite
Backend:	Node.js, Express.js, FastAPI, REST APIs, MVC Architecture, Async Programming
Databases:	MongoDB, MongoDB Atlas, MySQL, Redis, Neo4j, Pinecone
AI:	RAG Pipelines, LangChain, LLMs, Multi-Agent Systems, Prompt Engineering, Generative AI, MCP
Messaging:	Apache Kafka, Redis Pub/Sub, Event-Driven Architecture, Message Queues
Integrations:	WhatsApp Business API (WABA), Meta Business API, WhatsApp Cloud API, Webhooks, WebSocket, Socket.IO, Bulk Messaging
Security:	JWT, Bcrypt, RBAC, OAuth2, Helmet.js, API Authentication
Testing:	Jest, Playwright, Vitest, E2E Testing, Integration Testing, API Testing
DevOps:	Docker, Kubernetes, Git/GitHub, SSH, Linux, Bash, CI/CD, AWS, Vercel, Render, cPanel
Tools:	Postman, VS Code, GitHub Copilot,Claude,Cursor,Codex
Core CS:	Data Structures & Algorithms, OOP, DBMS, OS, Computer Networks, System Design

Projects

AI Project Estimator | FastAPI, React, Gemini AI, GitPython, Recharts, MongoDB Atlas, Multi-Agent Live Code

- Built a multi-agent AI estimation platform using FastAPI and Gemini AI with a Repo Intelligence module that analyzes GitHub repositories through GitPython.
- Generated automated cost, timeline, risk, and architecture reports; designed MongoDB Atlas APIs and a Recharts dashboard for estimation comparison.

Automated Bank Statement Processing | Python, Gmail API, PyPDF, BeautifulSoup4, CustomTkinter, PyInstaller, macOS launchd Code

- Automated bank statement processing for 10+ banks, reducing manual effort by **90–95%** through PDF extraction, classification, and workflow automation.
- Implemented SHA-256 duplicate detection and packaged the solution as a standalone desktop application with a macOS launchd monitoring daemon.

Studify Hub | JavaScript, Bootstrap 5, CodeMirror, Groq API, Web Speech API Live Code

- Built a student productivity platform with a seven-language CodeMirror editor, Groq AI assistant, code executor, translator, chat history, and Web Speech API text-to-speech.

Experience

Tech Intern | Droot Private Limited (Slnko Energy Pvt. Ltd.), Noida *July 2026 – Present*

- Designed a Knowledge Graph-augmented RAG system for Slnko Energy using LangChain, Neo4j, Pinecone, and LLMs for graph-aware semantic search, contextual retrieval, and intelligent AI responses.
- Developed WhatsApp Business integration for Slnko Energy Pvt. Ltd. with Meta Business API and WhatsApp Cloud API using secure webhooks for receiving, processing, and responding to customer messages.
- Implemented webhook-based interaction workflows, message event handling, API authentication, delivery-status tracking, and integration with internal business systems.
- Worked with REST APIs, Socket.IO, cloud deployment, SSH-based server management, Jest testing, and Playwright end-to-end testing.

Freelance Full-Stack Developer *Nov 2025 – Jan 2026*

Dasgupta Maiti & Associates, Kolkata Live

- Delivered production MERN platform for 100+ users: JWT/Bcrypt RBAC, Zoho OAuth2 mail, Cloudinary CDN, multi-collection search engine, and react-helmet-async SEO with XML sitemaps.
- Built custom `deploy_prep.sh` CI/CD pipeline for one-click cPanel builds; configured SSL, DNS (MX, SPF, DKIM), and pdf-parse data migration — achieving 99%+ uptime.

Yashoda Bhawan, Jharkhand Live

- Built a full-stack hostel management system using React, TypeScript, Vite, Tailwind CSS, Node.js, Express, MySQL, Sequelize, and Cloudinary.
- Developed student and administrator dashboards for registration, room allocation, attendance, events, gallery management, payments, and administrative operations.
- Implemented JWT authentication, RBAC, OTP email verification, password recovery, administrator approval workflows, invoice generation, and encrypted student information.

Frontend Developer Intern | Coding Bits *June 2025 – July 2025*

- Built reusable React component library used across 3+ features; delivered all sprints on time in Agile/Scrum with consistent peer code reviews.

Education

B.Tech – Information Technology, Bhagwan Parshuram Institute of Technology, GGSIPU, New Delhi 2023 – 2027
CGPA: **8.0**

Competitive Programming

Solved **300+ problems** on LeetCode & GeeksforGeeks in Arrays, Trees, Recursion, Sliding Window, Greedy Algorithms, and Dynamic Programming.

Hackathons

HackHazards 2025 (Team Codesync) | **Adobe India Hackathon** (Team Codesync) | **GDG Solution Challenge** | **L&T Createch 2026** Certificates

Certifications & Leadership

- **Udemy:** Full Stack AI Engineer 2026 – Generative AI & LLMs | Professional Certificate in DevOps | Java DSA + LeetCode Exercises | Full-Stack E-Commerce (Spring Boot & React) | CSS, JS, Bootstrap, Java & C++.
- **NSS Volunteer** – Community development programs, event organization & leadership All Certificates